PPP-L-1607A
January 19, 1971
SUPERSEDING
Int. Fed. Spec. PPP-L-001607(GSA-FSS)
March 5, 1970

FEDERAL SPECIFICATION

LAGGING, WOOD, WIREBOUND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for two types of pre-fabricated lagging, (see 6.1).

1.2 Classification. Lagging covered by this specification shall be of the following types as specified (see 6.1).

Type I - Outside (for wood reels only, see Fig. 1).
Type II - Inside (for wood or metal reels, see Fig. 1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specification, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specification required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies.)

Military Specification:

MIL-W-6110 - Wood, Determination of Moisture Content of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8115

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Nongovernmental:

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street N.W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Wood. Wood members used in the fabrication of lagging shall conform to the requirements of MIL-STD-731, except that only woods from Groups III and IV may be used.

3.1.1.1 Moisture content. The lagging boards shall have a moisture content at the time of fabrication not greater than 18 per cent, nor less than 9 per cent of their oven dry weight.

3.1.1.2 Knots. Knots, if any, shall be sound. The diameter of any knot in a lagging board shall not exceed 1/3 the width of the board. The distance from the outer edge of any knot to the nearest end of the board shall be not less than 2 inches.

3.1.1.3 Decay. Each piece of wood shall be free from decay.

3.1.1.4 Grain divergence. The divergence of the grain shall not exceed 1 inch in 8 inches of length.

3.1.1.5 Infestation. Boards shall be free from any worm or beetle infestation.

3.1.2 Wire. Binding wires shall be 13 gage (.091) diameter low carbon annealed steel wire. All wires shall have a galvanized coating and shall be smooth and shall not flake or peel when tested in accordance with 4.3.2. Wires shall be spaced from 4 to 6 inches in from ends of boards, be uniformly spaced across the width with the maximum spacing between wires not to exceed 12 inches.

3.1.3 Staples. All staples shall be a minimum 18 gage (.0475) diameter low carbon steel and shall have a galvanized coating. Galvanized coating shall be smooth and shall not flake or peel when tested in accordance with 4.3.2. The length of staples shall be as specified by Figure 1 for applicable board thickness. Staple spacing shall not exceed 1-3/4 inches. There shall be two staples driven astride each wire into each board and clinched.

3.1.4 Boards. Boards shall be resawn lumber, cut to length as specified (see 6.2), and the edges shall be straight and square with the surface of the lumber. The width of all boards shall be 2-5/8 inches. The thickness shall be as specified in Figure 1. A tolerance of plus or minus 1/16 inch shall apply to length, width and thickness specified. The maximum number of boards consistent with the lagging length specified (see 6.2) shall be used. The spacing between boards shall be approximately equal along the length of lagging specified. This spacing is normally 1/8 to 1/4 inch depending on the thickness of boards and length of lagging, but in no case shall the space between boards exceed 3/8 inch.

3.2 Dimensions.

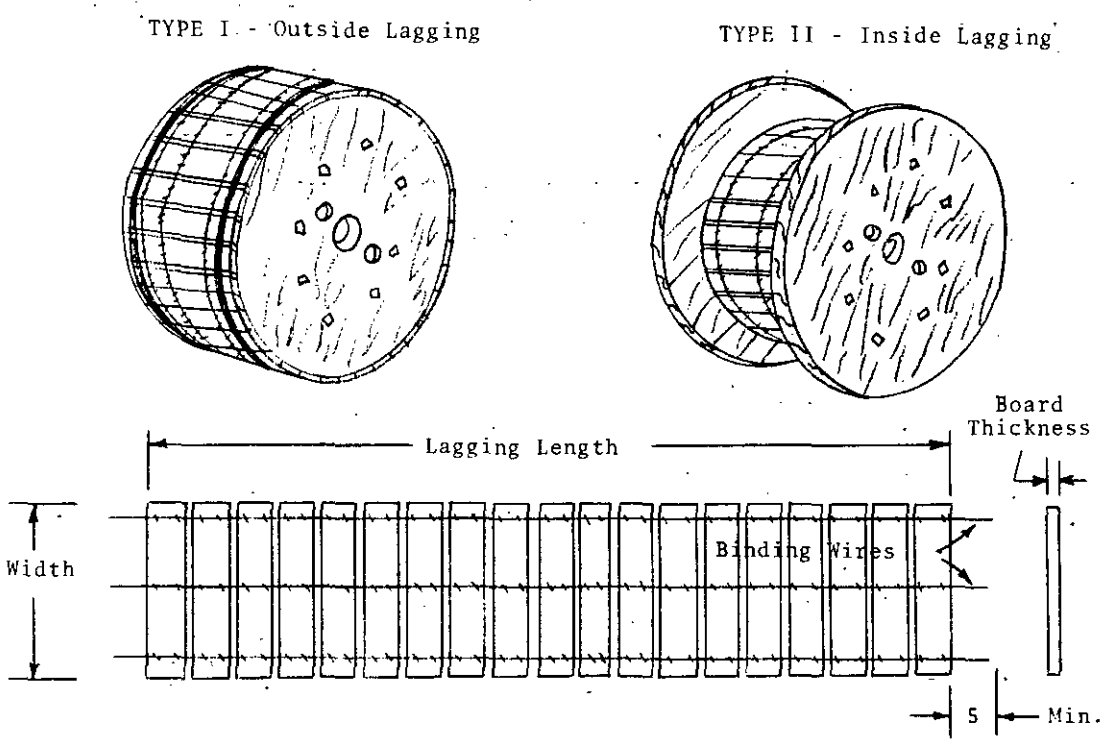
3.2.1 Lagging length. Type I outside lagging length shall equal the circumference of the reel expressed in inches rounded off to the closest 1/8 inch. Type II inside lagging length shall equal the circumference over the cable expressed in inches rounded off to the closest 1/8 inch.

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3.2.2 Lagging width. Type I outside lagging width shall equal the width of the reel minus 1/2 inch expressed in inches to the nearest 1/8 inch. Type II inside lagging width shall equal the traverse width of the reel minus 1/4 inch expressed in inches to the nearest 1/8 inch.

3.2.3 Wire length. The length of each binding wire shall be equal to the lagging length plus a minimum of 5 inches on each end.

3.3 Where this specification is referenced in documents in conjunction with reels for shipment of electric cable, cord or wire, the requirements of the appendix shall be a mandatory part of this specification.



Thickness of Boards Required

	Group III Woods	Staple Length*	Group IV Woods	Staple Length*
Type I - Outside lagging				
Width not exceeding 30 in.	1/2	3/4	3/8	9/16
Width over 30 inches	7/8	1	5/8	7/8
Type II - Inside lagging				
Width not exceeding 30 in.	3/8	9/16	1/4	7/16
Width over 30 inches	1/2	3/4	3/8	9/16

*All staples shall be 18 gage (minimum) - See paragraph 3.1.3.

FIGURE 1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections and tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of end item. Examination of the end items shall be in accordance with the classification of defects listed in 4.2.1 and MIL-STD-105 using inspection level S-2 and an AQL of 4.0 per cent defective for major defects and 6.5 per cent defective for minor defects.

4.2.1 Classification of defects. The sample unit shall be one completely assembled lagging.

TABLE I. Classification of defects

Defects	Major	Minor
Lagging length not as specified	X	
Lagging width not as specified	X	
Board thickness not as specified	X	
Decay in wood	X	
Broken Board	X	
Broken or missing wire	X	
Staple length not as specified	X	
Wire ends too short to make twist closure	X	
Worm or beetle infestation present	X	
Gage of wire not as specified		X
Wire not galvanized		X
Slope of grain exceeding one inch in eight inches of board length		X
Space between boards exceeds 3/8 inch		X
Knot exceeding 1/3 the board width		X
Wood group not as specified		X

4.3 Tests

4.3.1 Moisture content. The complete end item shall be tested in accordance with MIL-W-6110 except that two readings shall be taken from different boards of the item. Failure of the average of the two readings to meet the requirements shall class the item as defective.

4.3.2 Adherence of coating on galvanized wire. The following tests shall be performed to determine the adherence of zinc coating on galvanized wire (see 3.2.2). The wire shall be wrapped in a close helix for at least two complete turns, at a rate of not more than fifteen turns per minute, around a cylindrical mandrel 3/16 inch in diameter. After this operation, the coating shall not flake or be removable by rubbing the bare fingers over the wire on the mandrel.

5. PREPARATION FOR DELIVERY

5.1 Packing (level C). Lagging shall be packed to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. In addition to any special marking required by the contract or order, the shipping bundle shall be marked in accordance with Fed. Std. No. 123, or MIL-STD-129 as specified (see 6.2).

6. NOTES

6.1 Intended use. The prefabricated lagging covered by this specification is intended for domestic or export packing and shipment of electric cable, cord and wire on wood or metal reels not exceeding 60 inches in diameter or 42 inches in traverse width.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, required.
(c) Quantity required.
(d) Length and width required expressed in inches to the nearest 1/8 inch in order of length by width. See Figure 1.
(e) Marking in accordance with Fed. Std. No. 123 or MIL-STD-129 and any special marking.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 69

Review activities:

Army - SM
Navy - YD
Air Force - 69

User activities:

Navy - YD
Air Force - 69

Preparing activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

GSA-FSS

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APPENDIX

10. SCOPE

10. This appendix covers requirements for steel strapping, closure, nailing and waterproofing of the contents of the reels.

10.2 Waterproof paper shall be used under lagging as specified by applicable Cable, Cord or Wire specification.

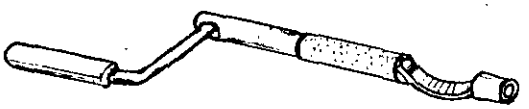
10.3 Type I, outside lagging shall be wrapped around the outside of the flanges and the binding wires shall be twisted to form a twist closure. Every sixth board of the lagging, including the first and last board shall be nailed to the wood flanges of the reel with cement coated nails. Strapping shall be as specified by applicable specification except that only two straps shall be required, applied over the flanges. Straps shall be stapled or nailed to the lagging at intervals of approximately 15 inches. See Figure 2 for one type of closure.

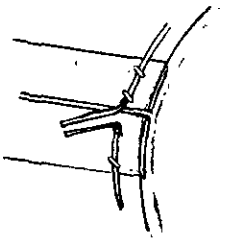
10.4 Type II, inside lagging shall be wrapped around the protected cable and the binding wires twisted to form a twist closure. See Figure 2 for one type of closure. Strapping, not normally required on inside lagging, shall be at user's option.

10.5 Strapping shall be performed by the packing activity. Strapping is not to be furnished by the lagging supplier unless specifically required by the contract or order.

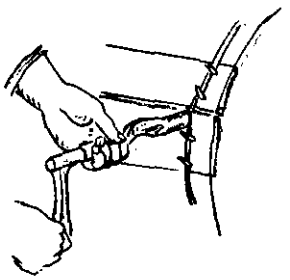
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Crank Twister

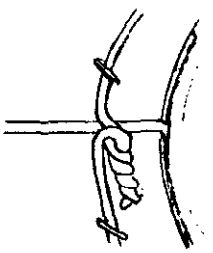




Bend wires at right angle to surface of board



Push end of bit over wires against board. Crank is rotated until wires cut off.



Knock twist against board parallel to the binding wire.

Wire twisting tools may be obtained from any wirebound lagging supplier, but wires may be twisted with any suitable tool to produce similar results.

FIGURE 2.

☆ U. S. GOVERNMENT PRINTING OFFICE : 1971 O - 436-361 (2320)

Orders for this publication are to be placed with General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification 10 cents each.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: General Services Administration, FSS (FMSX), Washington, D.C. 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

PPP-L-1607A - Lagging, Wood, Wirebound

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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